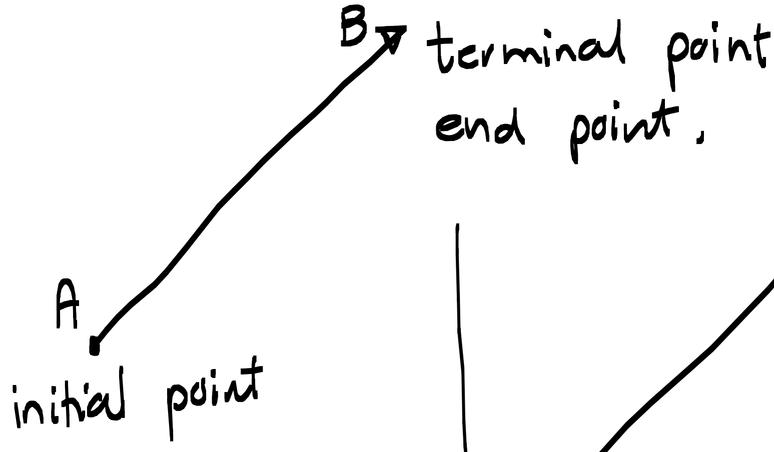
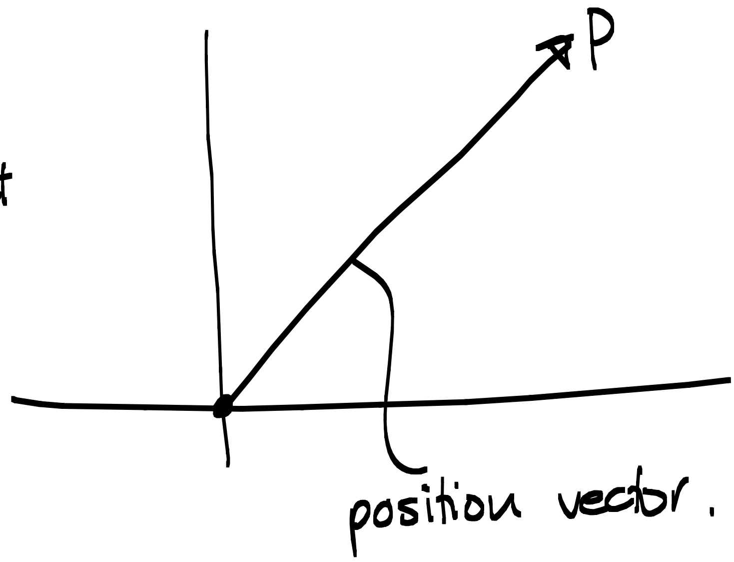
Unit 6·2 Vectors p 798

Note Title 2014/05/22

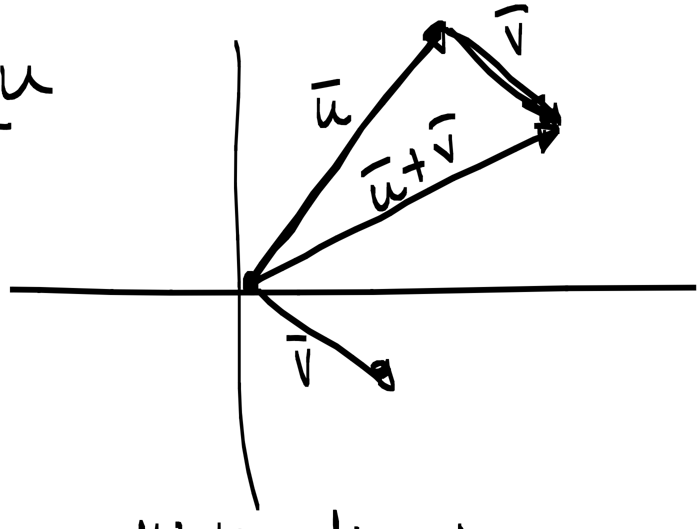
Vector – Quantity with both magnitude and direction.





Vector addition:

Substraction

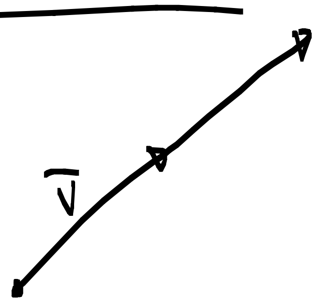


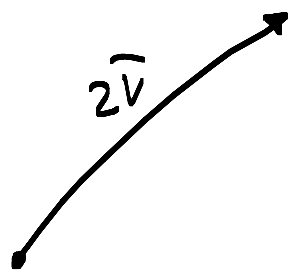
$\bar{v} \cdot \bar{u}$

dot product.

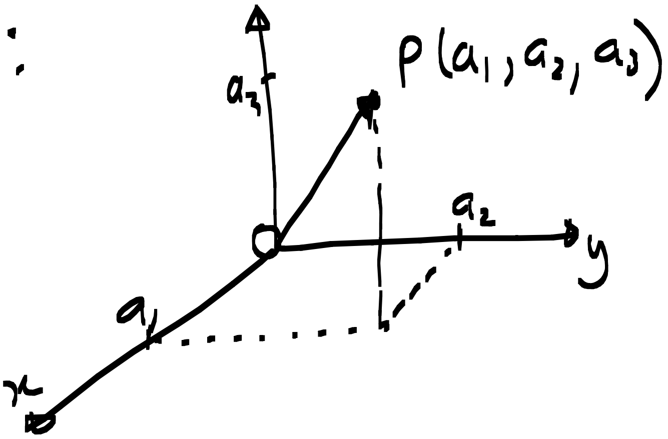
Scalar multiplication:

$2\bar{v}$





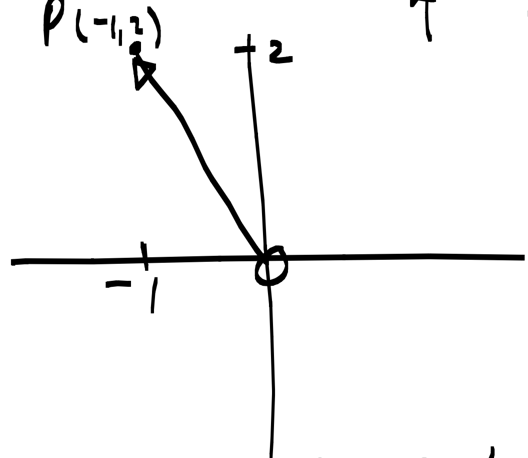
Components:



$P(a_1, a_2, a_3)$

Position vector $\overline{OP} = \langle a_1, a_2, a_3 \rangle$

Sketch $\langle -1, 2 \rangle$



Given $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ points in $\mathbb{R}^3$ then

$$\overrightarrow{AB} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle.$$

Vector add, subst, scalar X

$$\bar{a} + \bar{b} = \langle a_1, a_2, a_3 \rangle + \langle b_1, b_2, b_3 \rangle \quad (p799)$$
$$= \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

Example:

Find $2\overline{AB} + 3\overline{CA}$ if

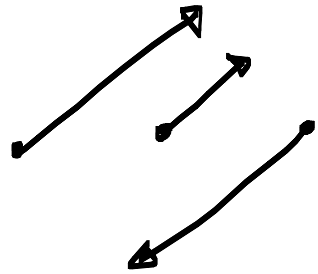
$A = (2, -1, 0)$
$B = (-3, 4, 1)$
$C = (1, 2, -2)$

$= 2\langle -5, 5, 1 \rangle + 3\langle 1, -3, 2 \rangle$

$= \langle -10, 10, 2\rangle + \langle 3, -9, 6\rangle$

$= \langle -7, 1, 8\rangle$ *

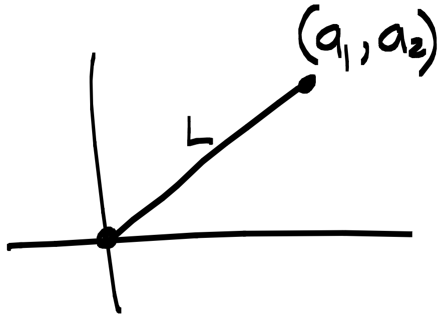
Vectors $\bar{a}$ and $\bar{b}$ are // when there exists a number $c \in \mathbb{R}$ so that.



$$\bar{a} = c\bar{b}$$

<u>Example:</u> If A(0,1) B(1,1) C(0,2) and D(3,2) use a sketch and the definition to prove that $\overline{AB} \,//\, \overline{CD}$. (SELF)

<u>Magnitude:</u> length of $\bar{v}$.



$$|\bar{a}| = \sqrt{a_1^2 + a_2^2}$$

$$|\bar{b}| = \sqrt{b_1^2 + b_2^2 + b_3^2}$$

$\boxed{\text{Unit vector}}$ - vector of length 1.

<u>Example:</u>

$\bar{a} = \langle 1, 1, 1\rangle$ ~ $|\bar{a}| = \sqrt{1^2 + 1^2 + 1^2}$

$= \sqrt{3}$ not a unit v.

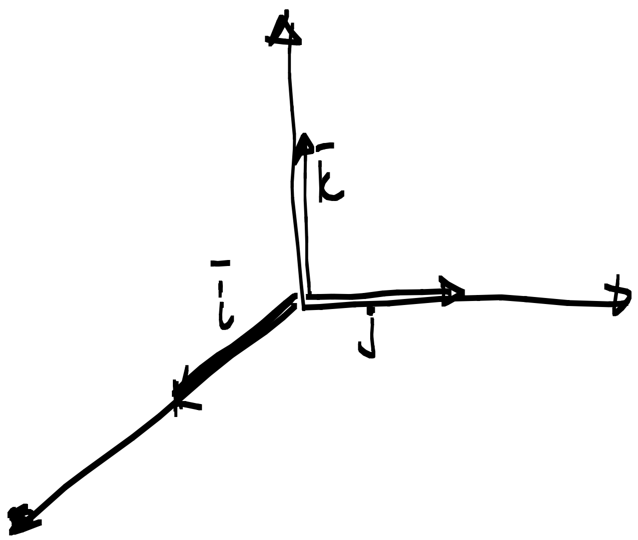
$\bar{b} = \langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\rangle$ ~ $|\bar{b}| = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1$ ✓

If $\bar{a} \neq 0$ then $\frac{\bar{a}}{|\bar{a}|}$ is a unit vector in the direction of $\bar{a}$.

Find the vector of length 2 in the opposite direction as $\bar{a} = \langle -3, 4 \rangle$.

$$-2\frac{\bar{a}}{|\bar{a}|} = -2\frac{\langle -3, 4\rangle}{\sqrt{9+16}} = -2\frac{\langle -3,4\rangle}{5}.$$

Standard basis vectors p796



$\bar{i} = \langle 1, 0, 0 \rangle$

$\bar{j} = \langle 0, 1, 0 \rangle$

$\bar{k} = \langle 0, 0, 1 \rangle$

$\langle 1,0,0\rangle + \langle 0,-2,0\rangle + \langle 0,0,6\rangle$

We write $\langle 1, -2, 6 \rangle = \bar{i} - 2\bar{j} + 6\bar{k}$

Find the unit vector in the direction of $2\bar{i} - \bar{j} - 2\bar{k}$. Magn. $\sqrt{4+1+4}$

$$\frac{2\bar{i} - \bar{j} - 2\bar{k}}{\sqrt{4+1+4}} = \frac{2\bar{i} - \bar{j} - 2\bar{k}}{3} = \frac{2}{3}\bar{i} - \frac{1}{3}\bar{j} - \frac{2}{3}\bar{k}$$

$$\frac{\bar{a}}{|\bar{a}|} = \frac{\langle 2, -1, -2 \rangle}{3} = \left\langle \frac{2}{3}, -\frac{1}{3}, -\frac{2}{3} \right\rangle$$